

일반물리실험II (BS106a-06)

학기	2026년 2학기
담당교수	박기성
이수구분	공통필수
학점	1.0
강의시간	월13:00-15:00(E2 - 220)
강의실	E2 219 (고전역학/전자기학 실험실)
소속	기초학부

교과개요

This course introduces experimental topics in electricity, magnetism, and optics. Students investigate electrostatic and magnetostatic phenomena, DC and AC circuits, and the properties of light as an electromagnetic wave. Through hands-on experiments and discussions, they develop an understanding of fundamental electromagnetic principles and their applications. The course culminates in creative team projects in which students design and conduct experiments on self-selected topics to explore innovative applications of electromagnetism.

교과목표

To gain practical insight into the fundamental concepts of electromagnetism underlying electrical engineering and optics through laboratory experiments.

To develop competence in collecting and analyzing experimental data to identify the physical principles governing observed phenomena.

To enhance the ability to design experiments, interpret results, and communicate scientific work effectively through written reports and presentations.

주차별 강의계획

1주차:Orientation

2주차:Equi-potential lines experiments

3주차:Ohm's law and DC RC circuits

4주차:Discussion and make-up lab

5주차:Wheatstone Bridge experiment

6주차:Solenoid Magnetic Field and the Principle of a Transformer

7주차:Discussion and make-up lab

8주차:Midterm Week (No Exam)

9주차:AC RLC Circuits

10주차:Brewster's Angle/Young's Double-Slit Interference

11주차:Discussion and make-up lab

12주차:Creative Lab 1

13주차:Creative Lab 2

14주차:Creative Lab 3

15주차:Creative Lab presentation

16주차:Final Week (No Exam)

수업방법

Students are expected to review the theoretical background related to each experiment in advance, and a quiz will be administered at the beginning of class.

Students work in teams to conduct assigned experiments, maintain laboratory notebooks, and submit laboratory reports within three days of completing the experiments.

Students collaborate in teams to design and carry out creative experiments and present their results to the class.

학습윤리

Students are expected to maintain academic integrity by clearly citing and properly acknowledging all references and sources used in the preparation of laboratory reports.

교과정책(방침)

Students are expected to actively participate in all laboratory activities.

Students are expected to actively engage in discussions and contribute to the exchange of ideas.

Students must comply with all requirements and guidelines related to personal hygiene and health.